



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Self-Calibrating Question Bank Engine Using Real-Time Learner Response Analytics

CSA0925 – Programming in Java

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Project Overview



Domain: Ed-Tech / Adaptive Learning / AI-Assisted Assessment

Our System:

Intelligent online question bank

Adapts question difficulty based on performance

Tracks **accuracy + time taken + topic performance**

Provides personalized practice and tests

Target Users:

Students / Learners

Administrators

Purpose:

Personalized learning instead of one-size-fits-all testing.



PROBLEM STATEMENT



The Problem

Traditional question banks provide:

- Same questions for every learner
- Fixed difficulty
- No consideration of solving time
- Limited performance insights
- Weak personalization

Impact

Students may practice too easy or too difficult questions without knowing what they actually need to improve.



MOTIVATION & NEED



Why This Project?

Students learn at different speeds and have different strengths.

Our system aims to:

- 🎯 Identify individual ability
- ⚡ Consider response time
- 📊 Detect weak topics
- 📈 Adapt question difficulty
- 🏆 Encourage continuous learning

Goal:

Make digital practice more personalized and measurable.

EXISTING SYSTEM / CURRENT PRACTICE

Existing Approach	Limitation
Static Question Bank	Same experience for everyone
Fixed Difficulty Tests	Cannot adapt to learner ability
Basic Online Quiz	Limited analytics
Manual Performance Review	Time-consuming
Generic Recommendations	Not personalized

Key Limitation

Most systems record answers — but do not continuously use them to improve the next question.

RESEARCH / KNOWLEDGE GAP

Paper	Method Used	Key Contribution	Research Gap
Dai et al. (2023)	Adaptive Testing + Recommendation	Used adaptive testing techniques for personalized learning recommendations.	Limited continuous question difficulty updates.
Liu et al. (2024)	Machine Learning + Computerized Adaptive Testing	Reviewed ML-based approaches for adaptive question selection.	Requires better real-time learner response integration.
Wang et al. (2024)	Deep Reinforcement Learning	Proposed intelligent question selection strategies for adaptive tests.	Focuses mainly on selection, not complete question bank calibration.
Sharpnack et al. (2024)	AutoML + Item Response Theory	Improved automatic item calibration using ML techniques.	Does not include personalized practice recommendation.

PROPOSED SOLUTION



A Self-Calibrating Question Bank with Adaptive Difficulty

-  Subject & topic-based learning
-  Adaptive question recommendation
-  Continuous performance analysis
-  Response-time tracking
-  Automatic difficulty calibration
-  Learner ability estimation
-  Personalized analytics
-  Practice, Test & Time-Attack modes

Core Idea: Answer → Analyze → Calibrate → Recommend → Repeat



OBJECTIVES



- Build an adaptive question bank
- Track accuracy and response time
- Dynamically estimate learner ability
- Automatically adjust question difficulty
- Visualize topic mastery
- Provide personalized recommendations
- Provide practice, test and time-attack modes
- Give admins question health & analytics



REQUIREMENTS



Functional

- Account creation & login
- Student dashboard
- Subject/topic selection
- Adaptive practice
- Tests
- Time Attack
- XP & streaks
- Skill Matrix
- Recommendations
- Admin analytics

Technical

Frontend: HTML/CSS/JavaScript + Thymeleaf

Backend: Java + Spring Boot

Database: MySQL

ORM: Spring Data JPA / Hibernate



TECHNOLOGY REVIEW



Frontend

HTML5 | CSS3 | JavaScript | Thymeleaf

Backend

Java | Spring Boot

Database

MySQL

Data Layer

Spring Data JPA | Hibernate

Development

IntelliJ IDEA | Maven

Architecture

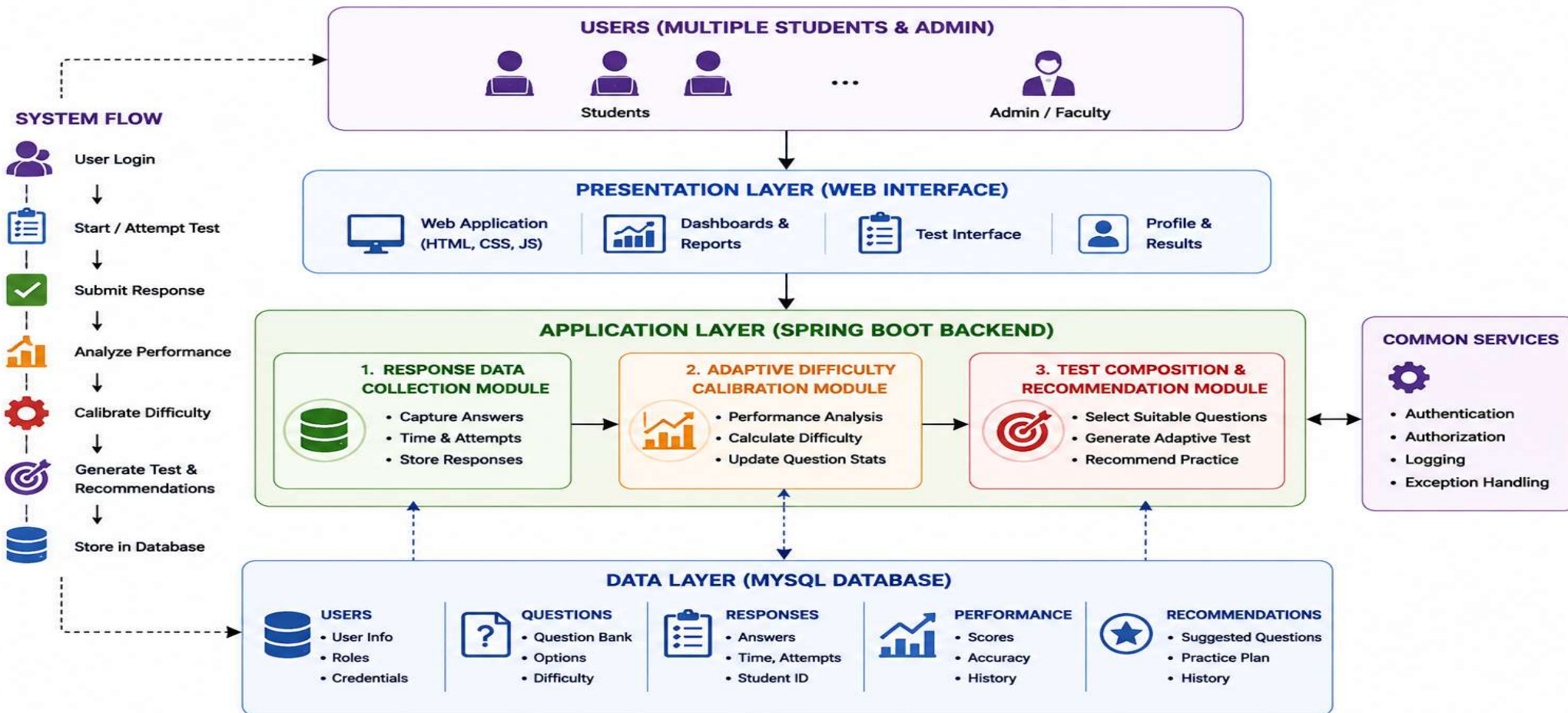
Browser → Spring Boot → Services → JPA → MySQL



Engineer to Excel



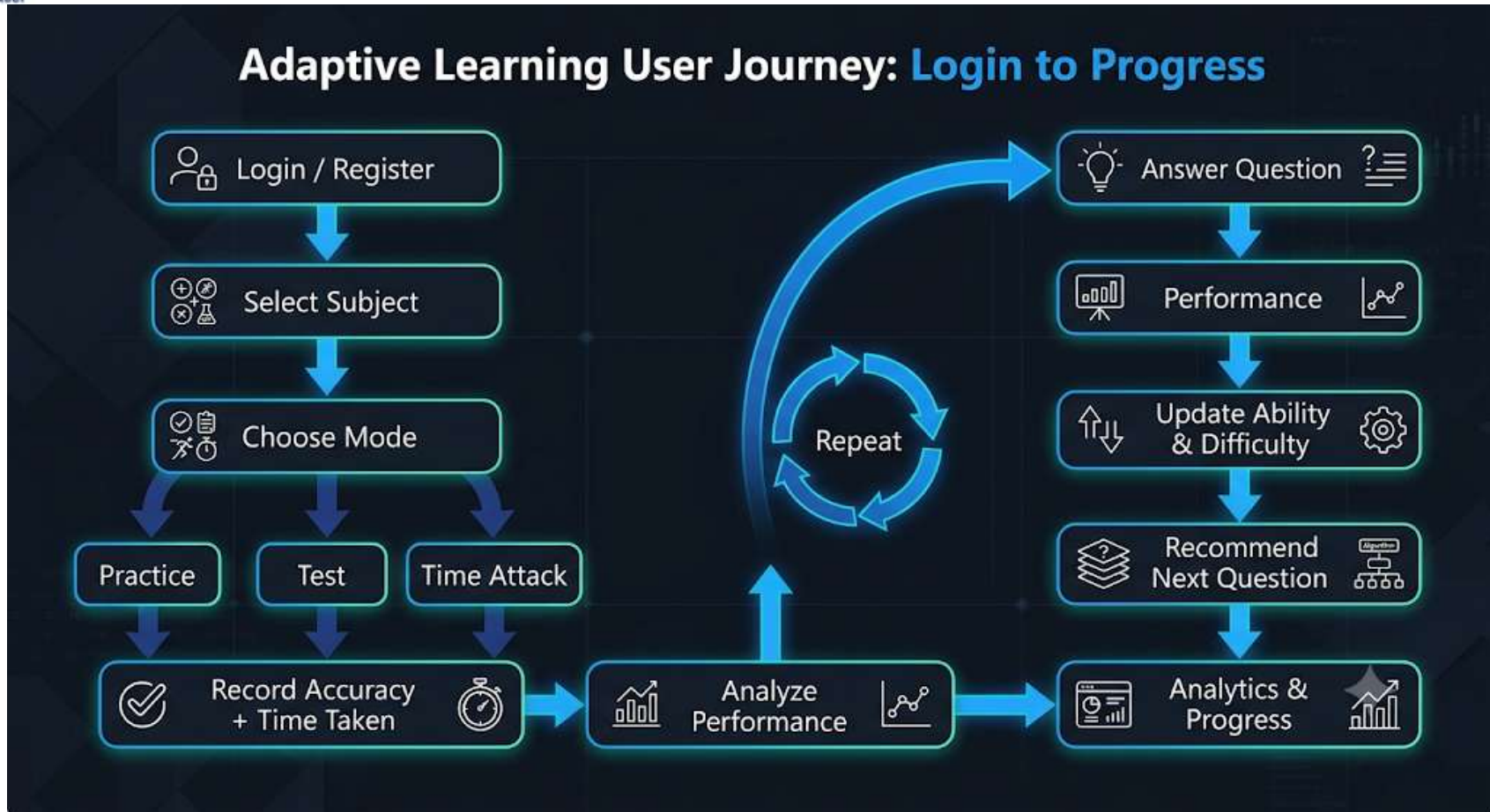
ARCHITECTURE





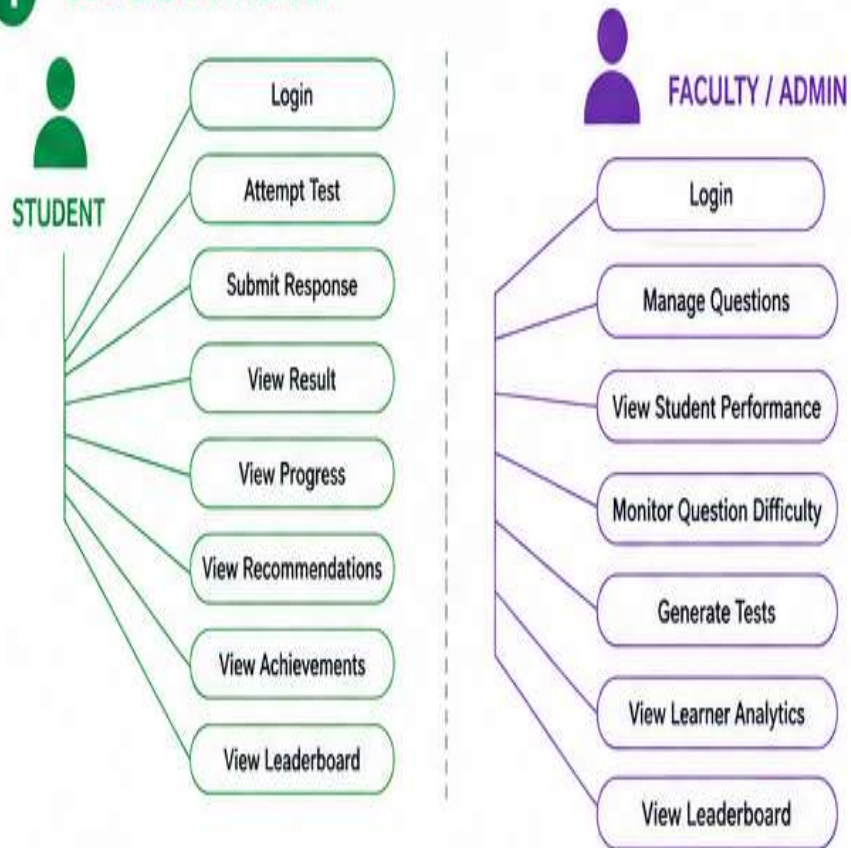
Engineer to Excel

WORKFLOW

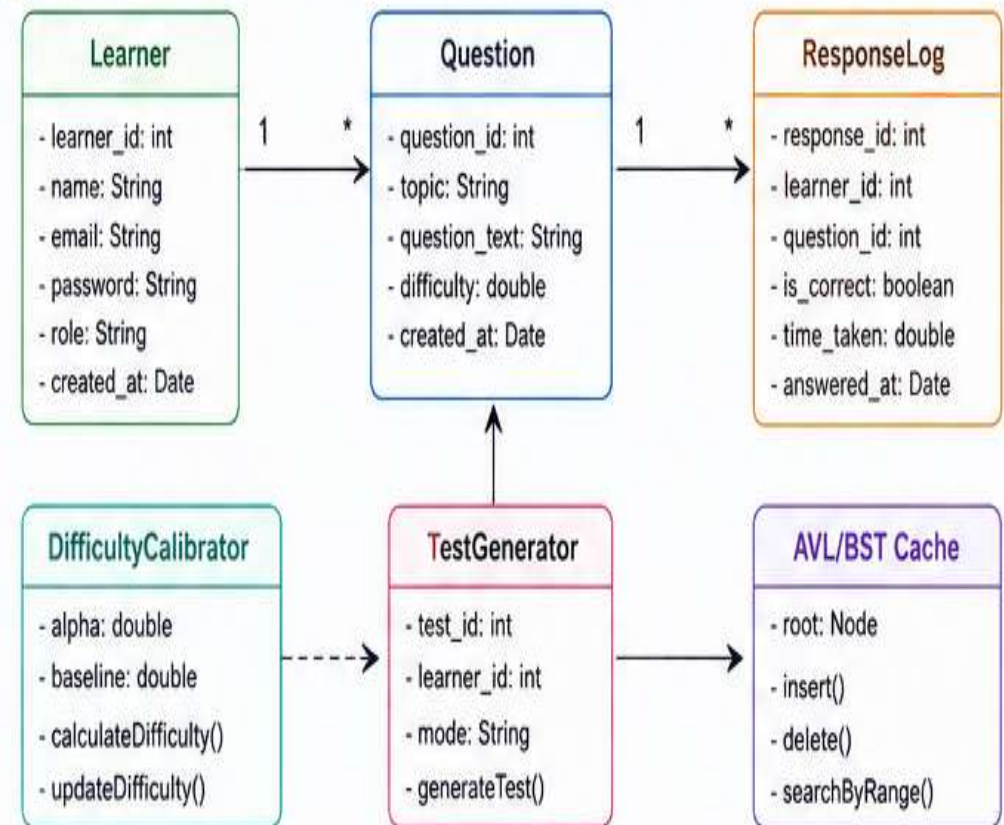


SYSTEM DESIGN

1 USE CASE DIAGRAM

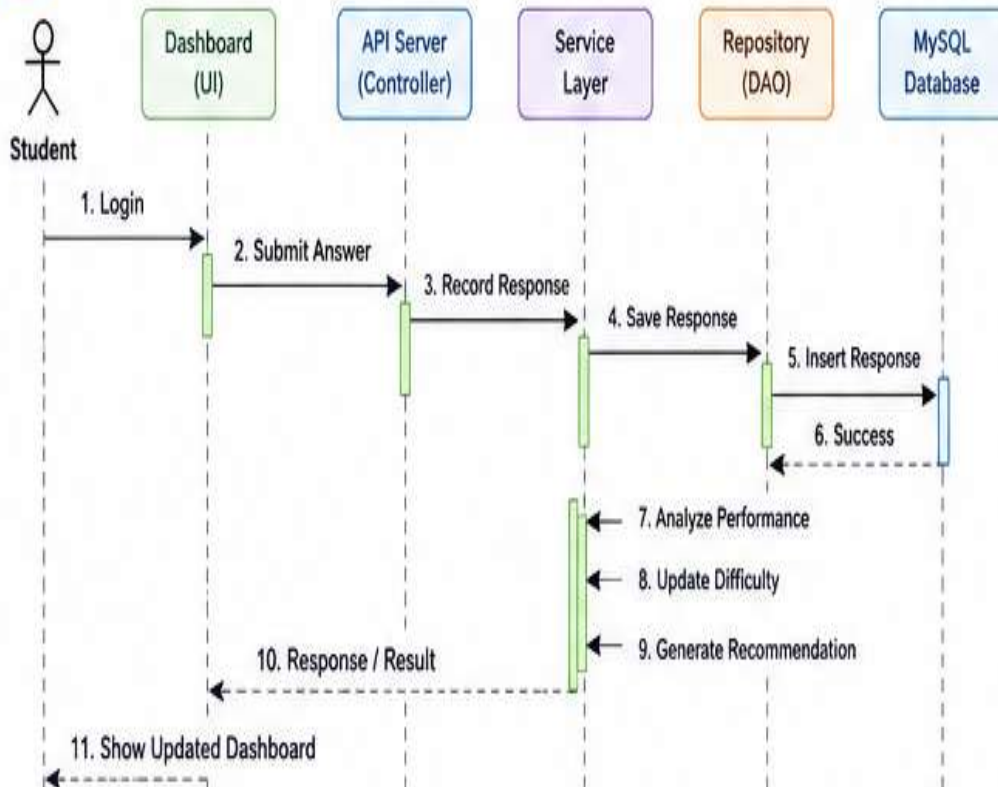


2 CLASS DIAGRAM

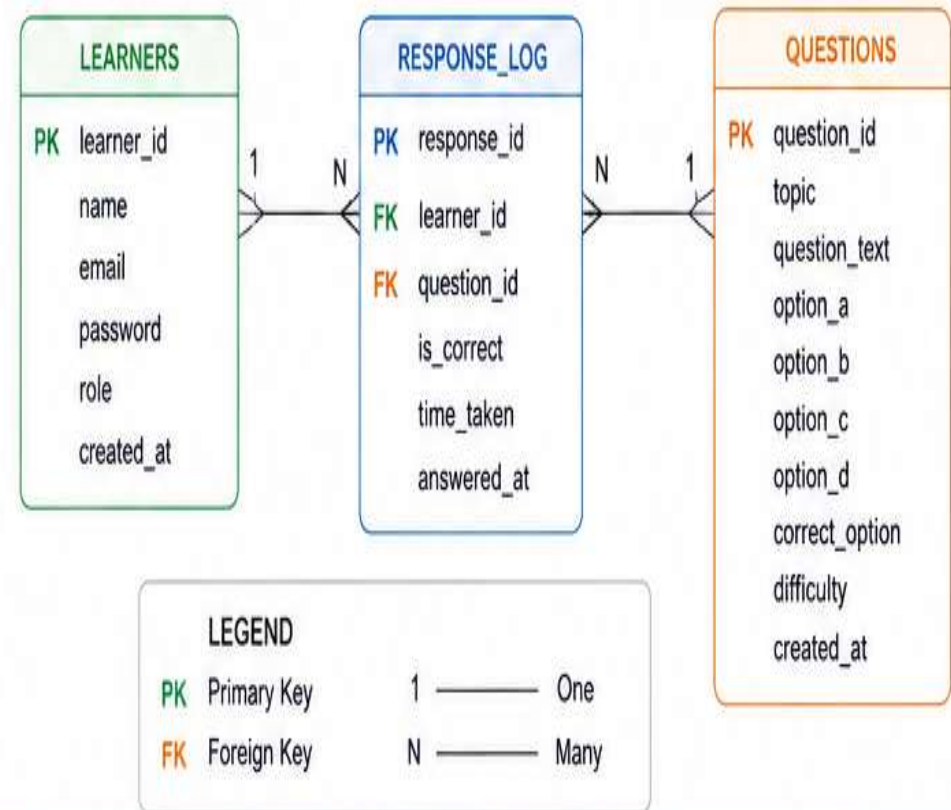


DATABASE DESIGN

3 SEQUENCE DIAGRAM



4 ER DIAGRAM



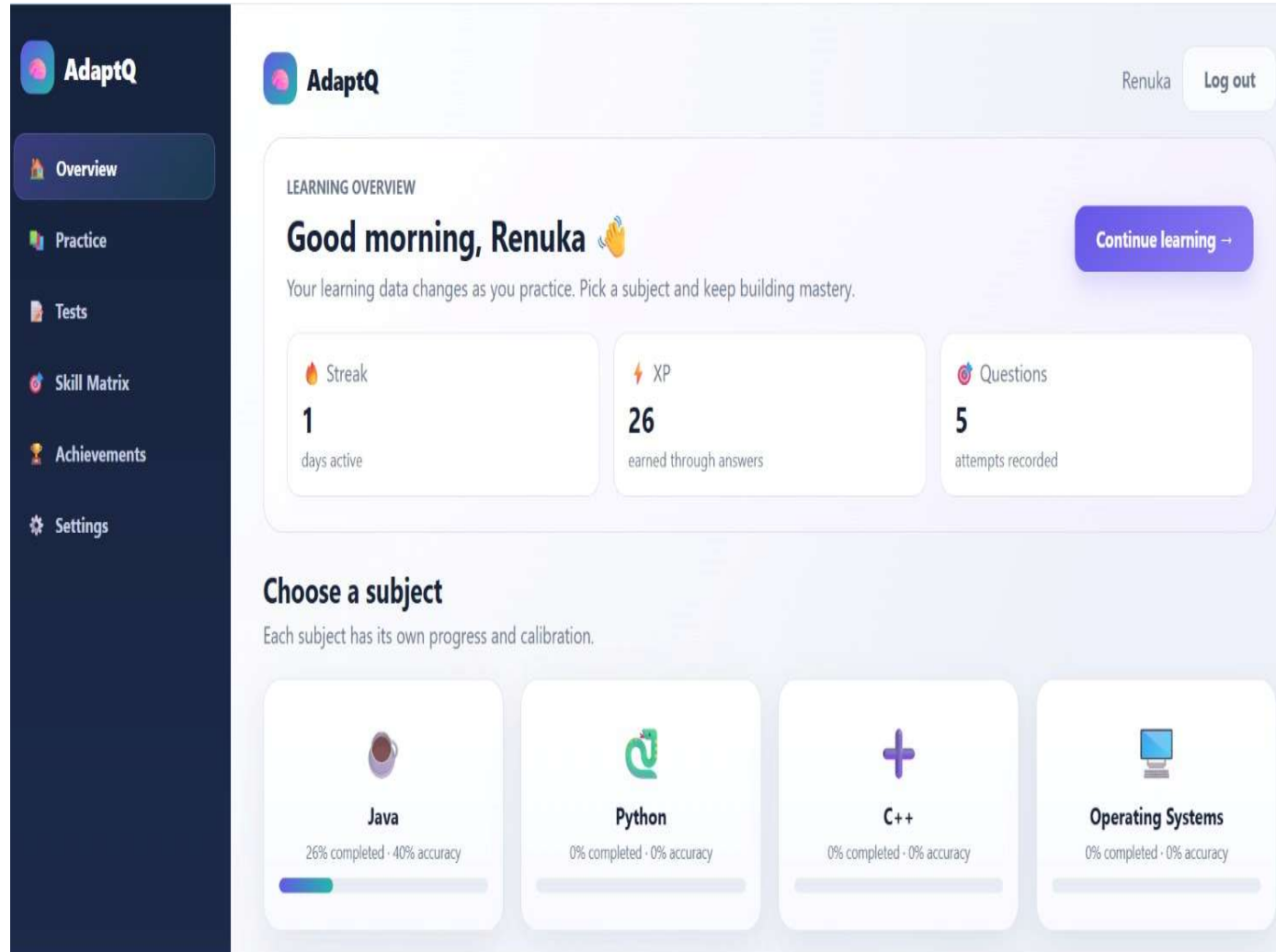
MODULE 1: RESPONSE DATA COLLECTION

Student Dashboard

Personalized Home

Shows:

-  Daily streak
-  XP & level
-  Accuracy
-  Average response time
-  Topic mastery
-  Continue Learning
-  Recommendations



The screenshot displays the AdaptQ student dashboard. On the left is a dark blue sidebar with the AdaptQ logo and navigation links: Overview, Practice, Tests, Skill Matrix, Achievements, and Settings. The main content area has a light blue header with the AdaptQ logo, the user's name 'Renuka', and a 'Log out' button. Below the header is a 'LEARNING OVERVIEW' section with a greeting 'Good morning, Renuka' and a 'Continue learning' button. It includes three summary cards: 'Streak' (1 day active), 'XP' (26 earned), and 'Questions' (5 attempts). The bottom section, 'Choose a subject', lists four subjects with their progress: Java (26% completed, 40% accuracy), Python (0% completed, 0% accuracy), C++ (0% completed, 0% accuracy), and Operating Systems (0% completed, 0% accuracy).

AdaptQ

Renuka [Log out](#)

LEARNING OVERVIEW

Good morning, Renuka 🖐️

[Continue learning →](#)

Your learning data changes as you practice. Pick a subject and keep building mastery.

Metric	Value	Unit / Description
Streak	1	days active
XP	26	earned through answers
Questions	5	attempts recorded

Choose a subject

Each subject has its own progress and calibration.

Subject	Progress
Java	26% completed · 40% accuracy
Python	0% completed · 0% accuracy
C++	0% completed · 0% accuracy
Operating Systems	0% completed · 0% accuracy

MODULE 2 : ADAPTIVE DIFFICULTY CALIBRATION

Adaptive Practice



Practice Flow

Subject → Topic → Difficulty → Question

During practice:

- Question displayed
- Timer starts
- Student answers
- Answer evaluated
- Time recorded
- XP awarded
- Ability updated
- Next question selected



Important

The next question is not randomly selected.
It is selected according to the learner's current performance.

The screenshot displays the AdaptQ application interface. On the left is a dark blue sidebar with the 'AdaptQ' logo and navigation links: Overview, Practice, Tests (highlighted), Skill Matrix, Achievements, and Settings. The main content area is titled 'Question-by-question review' and shows two questions. Question 1, 'Which built-in type stores key-value pairs?', is marked 'Correct' and has 'C. dict' as the selected answer. Below it is a hint: 'A dictionary stores data as key-value pairs.' and metadata: 'Data Structures', 'Difficulty: Medium (-0.07)', and '8.6s taken'. Question 2, 'Which symbol starts a single-line comment in Python?', is marked 'Incorrect' and has 'B. #' as the selected answer. Below it is a hint: 'Python uses # for single-line comments.'

MODULE 3: TEST COMPOSITION & RECOMMENDATION

Tests & Time Attack

Adaptive Test

Questions dynamically respond to performance.

Time Attack

Focuses on:

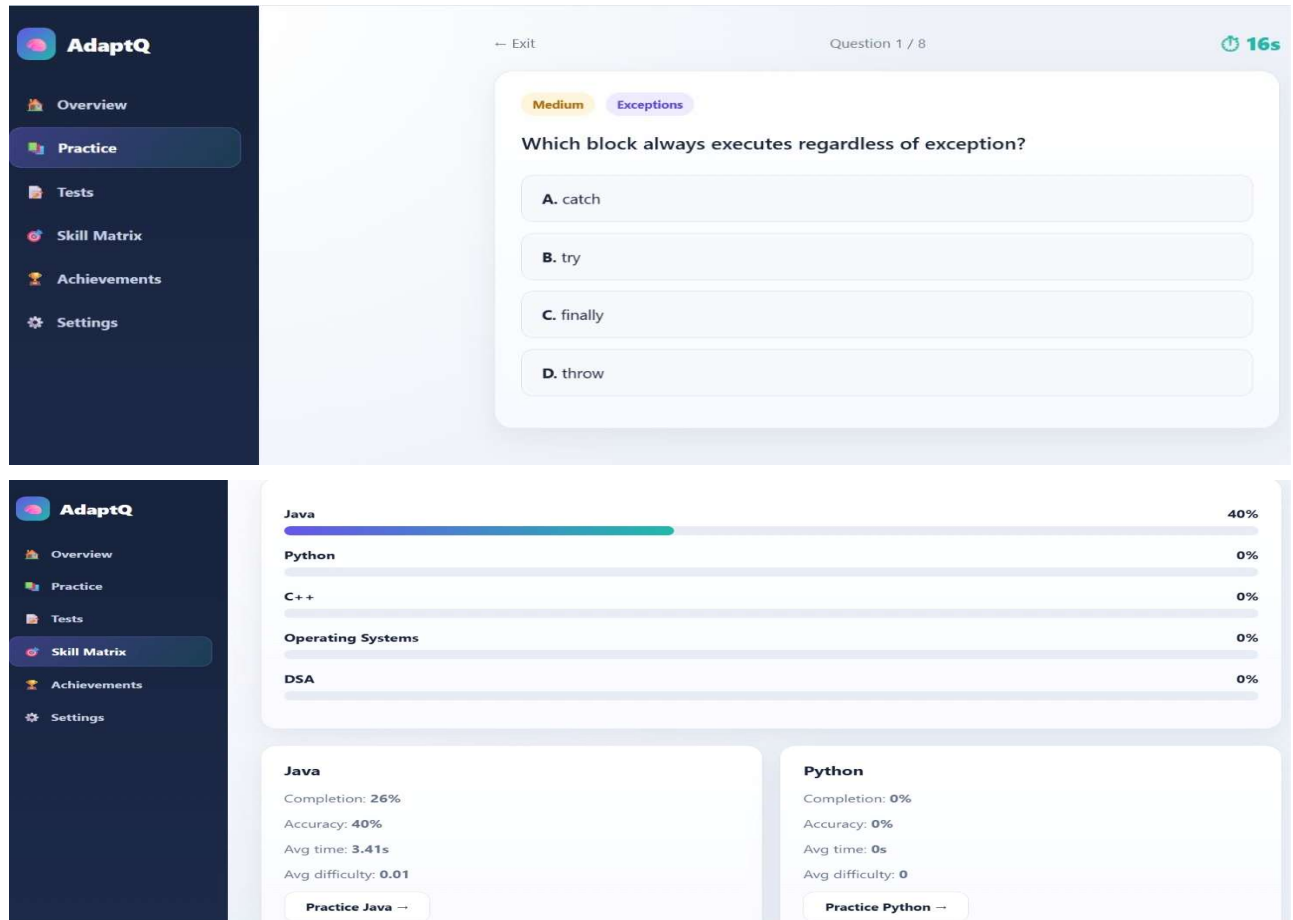
Questions the student can solve correctly but takes too long to answer.

Modes

Practice

Adaptive Test

Time Attack



The screenshot displays the AdaptQ application interface. On the left is a dark sidebar menu with the following options: Overview, Practice (highlighted), Tests, Skill Matrix, Achievements, and Settings. The main content area is divided into two sections. The top section shows a question card for 'Question 1 / 8' with a 'Medium' difficulty level and a timer of '16s'. The question is 'Which block always executes regardless of exception?' with four options: A. catch, B. try, C. finally, and D. throw. The bottom section shows a 'Skill Matrix' with progress bars for various topics: Java (40%), Python (0%), C++ (0%), Operating Systems (0%), and DSA (0%). Below the skill matrix, there are two panels for 'Java' and 'Python' showing completion and accuracy statistics. The Java panel shows: Completion: 26%, Accuracy: 40%, Avg time: 3.41s, Avg difficulty: 0.01, and a 'Practice Java' button. The Python panel shows: Completion: 0%, Accuracy: 0%, Avg time: 0s, Avg difficulty: 0, and a 'Practice Python' button.

Topic	Progress
Java	40%
Python	0%
C++	0%
Operating Systems	0%
DSA	0%

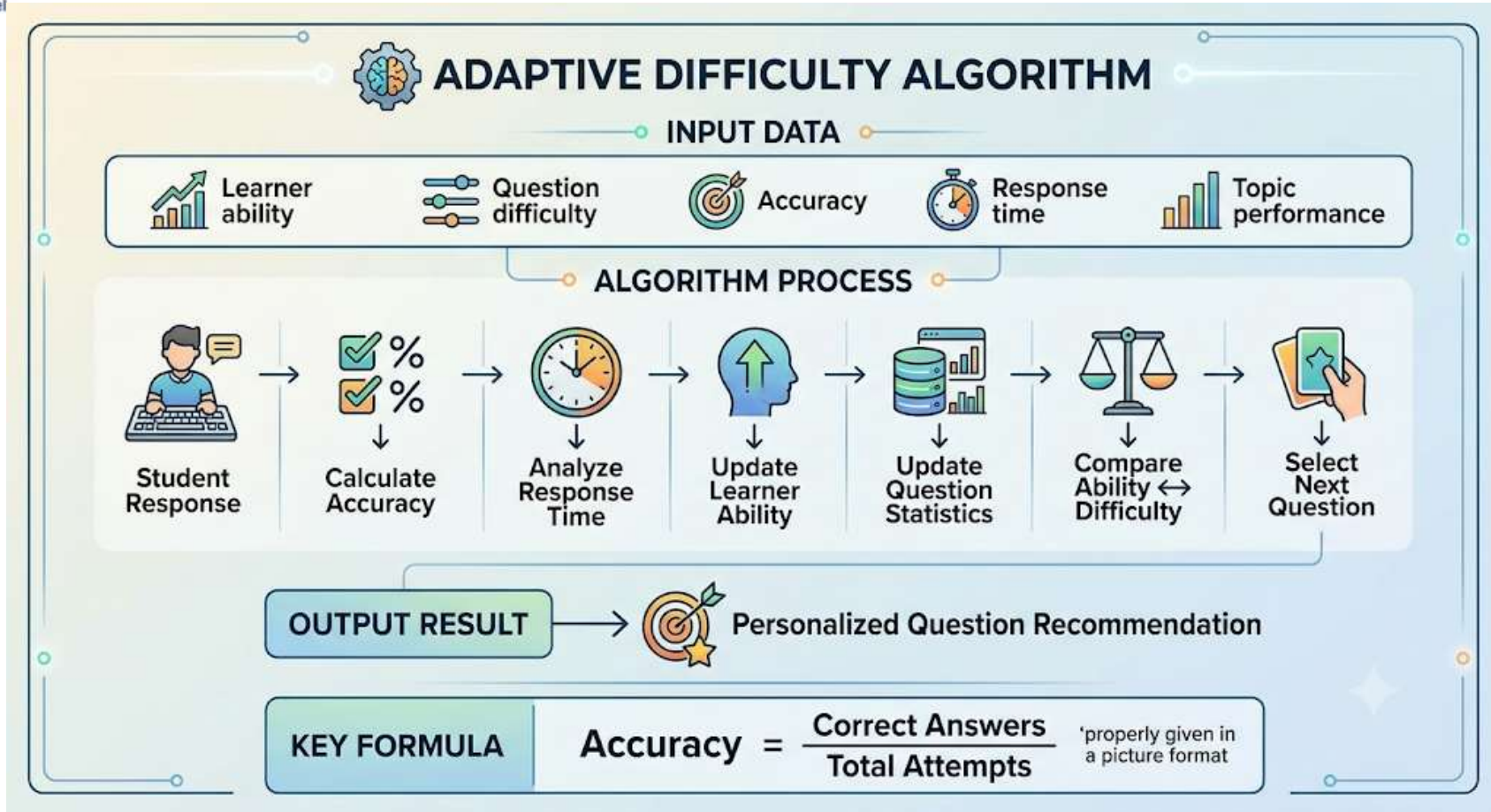
Topic	Completion	Accuracy	Avg time	Avg difficulty	Action
Java	26%	40%	3.41s	0.01	Practice Java
Python	0%	0%	0s	0	Practice Python



Engineer to Excel



ALGORITHM



RECOMMENDATION



Recommendation Engine

The system identifies:

-  Strong topics
-  Developing topics
-  Weak topics

Then recommends:

- **What to practice**
- **Which topic**
- **Which difficulty**
- **Which questions**

Example

Java → Exception Handling → Needs Practice

Instead of:

"Study Java."

PROTOTYPE



 AdaptQ
 [How it works](#)
[Features](#)
[Dashboard](#)
[Sign in](#)

SELF-CALIBRATING QUESTION BANK

Every answer makes your next question smarter.

AdaptQ combines adaptive difficulty, response-time analytics, skill tracking and personalized practice in one clean learning experience.

[Create your account →](#)
[▶ See how it works](#)

HOW IT WORKS

See the learning engine adapt in real time.

Answer → analyze → recalibrate → recommend.

1

AdaptQ / Student Home

LIVE ENGINE

 AdaptQ
 [How it works](#)
[Features](#)
[Dashboard](#)
[Sign in](#)

One place for your complete learning picture.

Progress, accuracy, speed and difficulty are visible instead of hidden behind static scores.



Performance graphs

Compare accuracy and average response time across subjects.



Subject mastery

Each subject has its own real progress based on your attempts.



Adaptive practice

Difficulty responds to your evolving ability.

CORE FEATURES

Built like a modern learning product.



Adaptive Practice

Targeted questions.



Time Attack

Speed-focused rounds.



XP & Achievements

Milestones and streaks.



Question Health

Automatic anomaly detection.

TESTING

Test	Expected
Registration	Account created
Login	Correct authentication
Question retrieval	Valid question
Answer submission	Correctness recorded
Timer	Response time recorded
Calibration	Difficulty updated
Analytics	Updated metrics
Recommendation	Relevant question

Result

Core workflow successfully integrated and tested.



RESULTS



- **Adaptive Learning** — Personalized question selection
- **Performance Tracking** — Accuracy + response time
- **Self-Calibration** — Learner ability & question difficulty
- **Visualization** — Topic mastery & performance graphs
- **Engagement** — XP, streaks & achievements
- **Analytics** — Detailed learner and question statistics

COMPARISON

Feature	Traditional	Q-Bank AI
Static questions	Yes	No
Adaptive difficulty	No	Yes
Response-time tracking	Limited	Yes
Topic mastery	Limited	Yes
Personalized recommendations	Limited	Yes
Time Attack	No	Yes
XP & streaks	Limited	Yes
Question health alerts	No	Yes



Our Advantage
 From "score reporting" → "learning intelligence".



ENGINEERING CHALLENGES



- **Challenge 1**

Difficulty calibration

→ Built a calibration service using learner responses.

- **Challenge 2**

Personalized question selection

→ Recommendation logic considers learner ability and topic performance.

- **Challenge 3**

Tracking response time

→ Capture question start/end timing.

- **Challenge 4**

Dynamic analytics

→ Backend APIs provide updated performance data.

- **Challenge 5**

Scalable question management

→ JPA repositories + MySQL persistence.

CONTRIBUTIONS

Student 1

-  Student UI/UX & Dashboard
-  Subject & Topic Selection
-  Adaptive Practice Flow
-  Time-Attack Mode
-  Performance Visualization
-  XP, Streak & Achievements
-  API Integration & Testing

Student 2

-  Backend Architecture
-  MySQL Database & JPA
-  Difficulty Calibration
-  Learner Ability & Recommendation
-  Analytics & Response Processing
-  Question Health & Admin APIs
-  API Integration & Testing

PROGRESS

Phase 1

- ✓ Requirement analysis

Phase 2

- ✓ Database & backend

Phase 3

- ✓ Authentication

Phase 4

- ✓ Adaptive engine

Phase 5

- ✓ Student dashboard

Phase 6

- ✓ Practice / Test modes

Phase 7

- ↻ Analytics refinement

Phase 8

- ↻ Deployment & validation

Use a **horizontal timeline** rather than a huge table.



LIMITATIONS

Current Limitations

- Current question dataset is limited
- Recommendation quality depends on available performance data
- Advanced AI explanation can be expanded further
- Large-scale deployment requires additional infrastructure

Important

These are opportunities for the next version, not failures of the prototype.

FUTURE

-  AI-powered personalized learning and explanations
-  Mobile application for learning on the go
-  Cloud deployment for large-scale users
-  Advanced predictive analytics for learner performance
-  More subjects and question types



OUTCOMES



Conclusion

We built a system that can:

- Personalize learning
- Track accuracy
- Track response time
- Adapt question difficulty
- Identify topic strengths/weaknesses
- Recommend practice
- Gamify learning
- Visualize performance
- Monitor question quality

Final Message

Every answer becomes data.

Every data point improves the next question.



Engineer to Excel



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A large, irregular splash of teal and blue watercolor paint serves as a background for the text. The colors vary in intensity, with darker blue in the center and lighter teal towards the edges, creating a soft, painterly effect.

Thank You